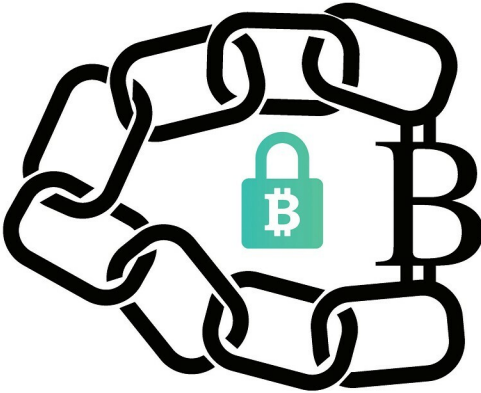


Blockchain-IoT based Management of Shared Resources between Trustless Parties

In this article, the authors discuss use cases for which blockchains and IoT devices can be used, instead of intermediaries, for Trustless parties to transact with each other. The authors also discuss the architecture and the open source technologies that were used in their solution.



Governments, communities or companies often need to track, control and manage assets and resources that they have a shared interest in. For example, in the logistics industry, a perishable item might get transported from a sender to a receiver. While in transit, the perishable item may have to be kept under a certain temperature or pressure. The item may get transported through multiple modes such as ship, train, truck, etc, and multiple transport operators might be involved before it reaches its destination. Payments have to be released to these parties if and only if they transported the item as per the contract.

Often, there is no trust between the sender, receiver and the various parties involved. This means that the parameters required to preserve a perishable item have to be tracked and controlled. And there needs to be a verifiable record of this tracking/controlling throughout the transportation period.

There is also a need for a trusted way to settle payments between the various parties once the shipment has been delivered as per the contract.

Another use case is in the energy sector, where smart grids are used for the distribution and consumption of electric power. There is a need for power generators and distributors to track the supply and distribution of power, and keep records of it, to eventually reconcile and settle accounts between the producers and distributors of power. A third use case could be to manage water resources stored in dams and shared between countries/states.

The list of such use cases is endless. Often these parties do not trust each other and end up appointing an intermediary third-party trusted by all of them to manage, control or track the resources, on their behalf. The intermediaries, however, may charge exorbitant fees for their services or may become